Module 5: Continuous RV's and their Distributions

5.1 Basic Definition (*Weiss §8.1*)

Definition 5.1. Continuous Random Variable

A random variable is called a **continuous random variable** if

$$\mathbb{P}(X = x) = 0 \qquad \text{for all } x \in R_X.$$

► Notes:

- The range R_X of a continuous random variable X is infinite and uncountable. (Otherwise, X would be discrete!)

  In other words, R_X is an interval of the real line or the union of many such intervals.

- Classical examples are: heights, weights, waiting times, etc. (measurements of some kind).

- Unlike what we did in the discrete setup, for any given value $x \in R_X$, the probability that $X = x$ is not what we will be interested in. (It is zero!)

  Rather, we will be interested in the probability that X is *close* to x, that is, something like

$$\mathbb{P}(x - h \leq X \leq x + h) \quad \text{or} \quad \mathbb{P}(x \leq X \leq x + h) \qquad \text{for some small } h > 0.$$

- Some random variables are not discrete nor continuous.

5.2 Cumulative Distribution Functions (*Weiss §8.2*)

Definition 5.2. Cumulative Distribution Function

The **cumulative distribution function** (CDF) of a random variable X, denoted F_X, is the real-valued function of $\mathbb{R} \to [0, 1]$ defined by

$$F_X(x) = \mathbb{P}(X \leq x) \qquad \text{for all } x \in \mathbb{R}.$$

The CDF is sometimes also referred to as **distribution function** or **cumulative distribution**.

► Notes:

- The CDF of a random variable is well defined for any type of random variable (discrete, continuous or other).

- The CDF fully determines the probability distribution of a random variable.

Example 5.1. CDF of a discrete RV

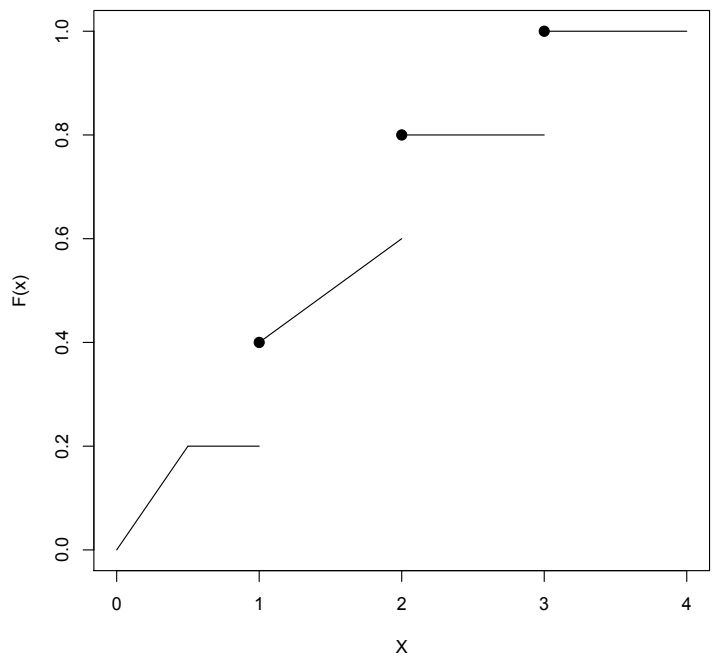
A discrete random variable Y has the following PMF

y	0	1	2	4
$P_Y(y)$	1/10	3/10	2/10	4/10

Obtain and graph the CDF of Y.

Example 5.2. A given CDF

Assume the random variable X has CDF F_X as shown on the next graph.



Calculate the following probabilities:

- $\mathbb{P}(X = 3)$
- $\mathbb{P}(X \leq 3/2)$
- $\mathbb{P}(X < 2)$
- $\mathbb{P}(2 < X \leq 3)$

Proposition 5.3. Basic Properties of CDF's

The CDF F_X of a random variable X satisfies the following properties.

1. F_X is a nondecreasing function.
2. F_X is everywhere right-continuous.
3. $\lim_{x \to -\infty} F_X(x) = 0$.
4. $\lim_{x \to \infty} F_X(x) = 1$.

Proposition 5.4. Basic Properties of CDF's

Let X be a random variable with CDF F_X. Then, for any $a < b$, we have that

1. $\mathbb{P}(a < X \leq b) = F_X(b) - F_X(a)$,
2. $\mathbb{P}(a \leq X \leq b) = F_X(b) - F_X(a-)$,
3. $\mathbb{P}(a < X < b) = F_X(b-) - F_X(a)$,
4. $\mathbb{P}(a \leq X < b) = F_X(b-) - F_X(a-)$,

where $F_X(a-) = \lim_{x \to a^-} F_X(x)$ denotes the left-hand limit of F_X at $x = a$.

Note that $F_X(a-) = \mathbb{P}(X < a)$.

Proposition 5.5. A Characterization of Continuous RV's

A random variable is continuous if and only if its CDF is a continuous function.

Example 5.3. Archery

An archer fires an arrow at a big circular target with a 1 m radius. Assume that the arrow will hit the target and that it is equally likely to hit any location on the target. Finally, let X denote the distance (in m) between the point of impact and the centre of the target.

Find the CDF of X. Also, calculate $\mathbb{P}(1/4 \leq X \leq 3/4)$.

5.3 Probability Density Functions (*Weiss §8.3*)

First, a continuous RV X takes a value *close* to x with probability

$$\mathbb{P}(x \leq X \leq x + h) \simeq hF'_X(x) \qquad \text{for very small } h > 0,$$

when F_X is differentiable at x, that is, this probability is proportional to $F'_X(x)$.

In other words, even though $F'_X(x)$ is not interpreted as a probability, we have that

- X is more likely to take a value close to x if $F'_X(x)$ is large,
- X is *not* likely to take a value close to x if $F'_X(x)$ is small.

Also, if F_X is differentiable, then

$$\int_a^b F'_X(x)\,dx = F_X(b) - F_X(a) = \mathbb{P}(a < X \leq b),$$

from the fundamental theorem of calculus.

Hence, in the continuous case, the derivative F'_X of F_X plays a role similar to a PMF!

Definition 5.6. Probability Density Function

Let X be a continuous random variable. Then, a nonnegative function f_X is said to be a **probability density function** (PDF) for X if

$$\mathbb{P}(a < X \leq b) = \int_a^b f_X(x)\,dx \qquad \text{for any } a < b \in \mathbb{R}.$$

A PDF is often referred to as a **density function**.

Proposition 5.7. Finding a PDF

Let X be a continuous random variable with CDF F_X. Then, X has a PDF if and only if there is a nonnegative function f such that

$$F_X(x) = \int_{-\infty}^x f(t)\,dt \qquad \text{for any } x \in \mathbb{R}.$$

In this case, f is a PDF of X and

$$F_X'(x) = f(x)$$

at all points where F_X is differentiable (or, equivalently, at all points where f is continuous).

Example 5.4. (refer to Example 5.3)

We have found that the CDF of X is given by

$$F_X(x) = \begin{cases} 0 & \text{if } x < 0, \\ x^2 & \text{if } 0 \leq x \leq 1, \\ 1 & \text{if } x > 1. \end{cases}$$

Find a PDF of X.

Proposition 5.8. Basic Properties of PDF's

The PDF f_X of a continuous random variable X satisfies the following properties.

1. $f_X(x) \geq 0$ for all $x \in \mathbb{R}$.
2. $\int_{\mathbb{R}} f_X(x)\,dx = 1$.

Alternatively, the second property can be written as

$$\int_{R_X} f_X(x)\,dx = 1,$$

where R_X denotes the range of X.

Example 5.5.

Assume that a random variable X has PDF

$$f_X(x) = \begin{cases} cx^3 & \text{if } 0 \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Find the value(s) of c for which f_X is a genuine PDF.

Example 5.6. Density function or not?

Let

$$f(x) = \begin{cases} \frac{3}{2}(x^2 - 1) & \text{if } 0 \leq x \leq 2, \\ 0 & \text{otherwise,} \end{cases} \qquad \text{and} \qquad g(x) = \begin{cases} 2e^{-2x} & \text{if } x > 0, \\ 0 & \text{otherwise.} \end{cases}$$

Are these functions genuine PDF's?

Proposition 5.9. Fundamental Probability Formula: Continuous case

Let X be a continuous random variable with PDF f_X. Then, for any $A \subset \mathbb{R}$, we have

$$\mathbb{P}(X \in A) = \int_A f_X(x)\, dx.$$

Example 5.7. (refer to Example 5.5)

Find $\mathbb{P}(X > 1/2)$ directly and by first obtaining the CDF of X.

Example 5.8. Waiting for the bus

Buses arrive at a bus stop at 15-minute intervals starting at 7 am. Assume that a passenger arrives at the bus stop at a random time X (given in minutes after 7 am) with PDF

$$f_X(x) = \begin{cases} x(60 - x)/36000 & \text{if } 0 \leq x \leq 60, \\ 0 & \text{otherwise.} \end{cases}$$

Find the probability that

- the passenger waits more than 10 minutes for the bus,
- the passenger waits no more than 5 minutes for the bus,
- on every day of a week, the passenger waits no more than 5 minutes for the bus .

5.4 Important Continuous RV's

5.4.1 Uniform RV's (*Weiss §8.4*)

Definition 5.10. Uniform Random Variable

A continuous random variable X is said to have the **uniform distribution** on the interval (a, b) if its PDF is given by

$$f_X(x) = \begin{cases} \dfrac{1}{b-a} & \text{for } a < x < b, \\ 0 & \text{otherwise.} \end{cases}$$

In this case, we write $X \sim \mathcal{U}(a, b)$ or $X \sim \text{Unif}(a, b)$.

Intuitively, X is then equally likely to lie anywhere in the interval (a, b).

Proposition 5.11. CDF of Uniform RV's

The CDF of $X \sim \text{Unif}(a, b)$ is given by

$$F_X(x) = \begin{cases} 0 & x \leq a, \\ \dfrac{x-a}{b-a} & a < x < b, \\ 1 & x \geq b. \end{cases}$$

Example 5.9. Point on a segment

A point X is selected at random (all points being considered equally likely) on a segment of length L. Calculate the probability that the sub-segments located to the left and to the right of X are both longer than $L/3$.

Example 5.10. Roots of a degree two polynomial

Suppose that $Y \sim \text{Unif}(-2, 4)$. Find the probability that the quadratic equation

$$x^2 + Yx + \frac{1}{4} = 0$$

has at least one real root.

5.4.2 Exponential RV's (*Weiss §8.4*)

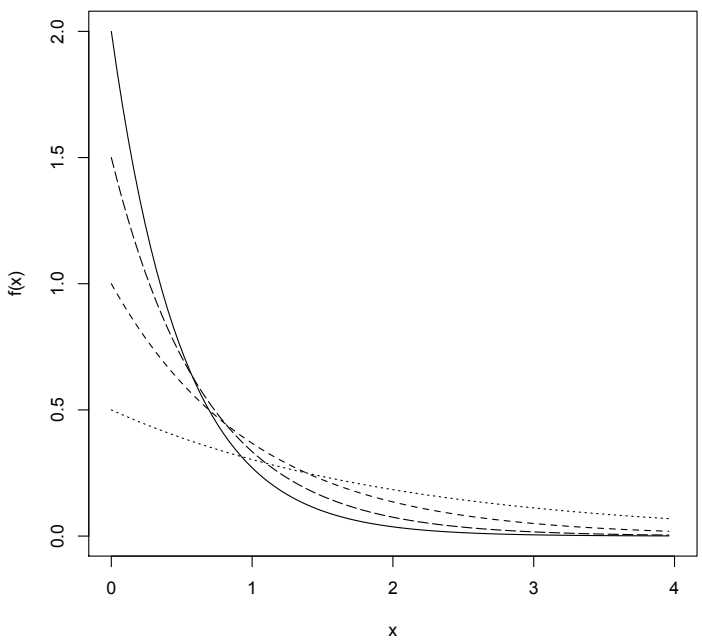
Definition 5.12. Exponential Random Variable

A continuous random variable X is said to have the **exponential distribution** with parameter $\lambda > 0$ if its PDF is given by

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x > 0, \\ 0 & \text{otherwise.} \end{cases}$$

We write $X \sim \mathcal{E}(\lambda)$ or $X \sim \text{Exp}(\lambda)$.

Exponential PDF's for $\lambda = 0.5, 1, 1.5, 2$.



Example 5.11. Link between Poisson and exponential distributions

For any $t > 0$, define $N(t)$ as the random number of occurrences of some event over the time interval $[0, t]$. In addition, assume that

$$N(t) \sim \text{Poisson}(\lambda t),$$

for some $\lambda > 0$. Show that the time T of occurrence of the first event (or, waiting time until the first occurrence of an event) has an exponential distribution with parameter λ.

Hint: Note that $N(t) = 0 \iff T > t$.

Proposition 5.13. CDF of Exponential RV's

The CDF of $X \sim \text{Exp}(\lambda)$ is given by

$$F_X(x) = \begin{cases} 0 & \text{for } x \leq 0, \\ 1 - e^{-\lambda x} & \text{for } x > 0. \end{cases}$$

Example 5.12. At the Pizzeria

Assume that, at Luigi's Pizzeria, the time in minutes between orders of pizza is exponentially distributed with rate $\lambda = 0.2$ orders per minute (that is, one order every 5 minutes).

Find the probability that

- no customer orders pizza in the next 5 minutes,
- the next order is placed between 10 and 15 minutes from now.

Proposition 5.14. Lack-of-memory Property of Exponential RV's

A positive continuous random variable X is said to have the **lack-of-memory property** if

$$\mathbb{P}(X > s + t \,|\, X > t) = \mathbb{P}(X > s) \qquad \text{for } s, t > 0.$$

The only continuous RV's with the lack-of-memory property are the exponential random variables.

In other words, a continuous random variable has the lack-of-memory property if and only if it has some exponential distribution.

Example 5.12. (cont'd)

Assuming that no order will come in over the next 10 minutes, find the probability that the next order is placed at least 15 minutes from now.

Luigi tells you that 90% of the times, he waits less than t minutes between orders. Find the value of t.

5.4.3 Normal RV's (*Weiss §8.5*)

Definition 5.15. Normal Random Variable

A continuous random variable X is said to have the **normal distribution** with parameters $\mu \in \mathbb{R}$ and $\sigma > 0$ if its PDF is given by

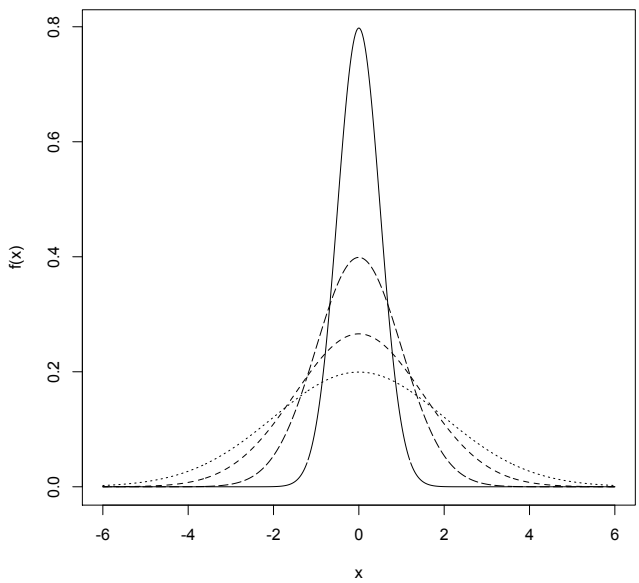
$$f_X(x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-(x-\mu)^2/2\sigma^2} \qquad \text{for } x \in \mathbb{R}.$$

We write $X \sim \mathcal{N}(\mu, \sigma^2)$ or $X \sim N(\mu, \sigma^2)$. Some textbooks use instead $X \sim N(\mu, \sigma)$.

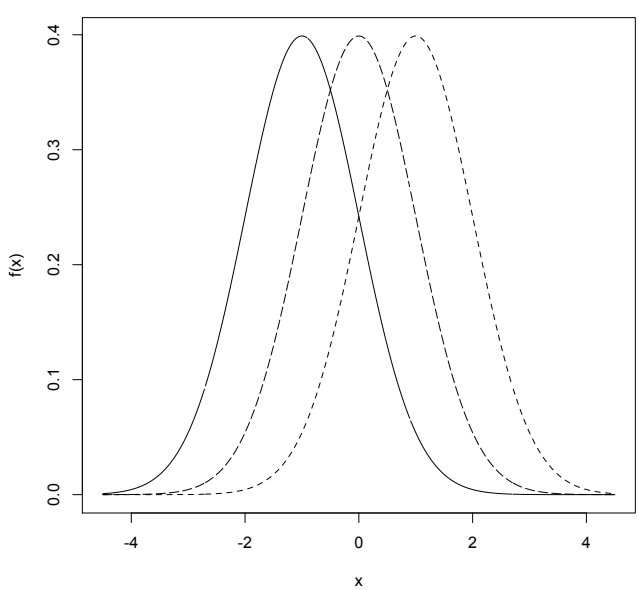
The normal distribution with $\mu = 0$ and $\sigma = 1$ is the **standard normal distribution**.

The normal distribution is also sometimes referred to as the **Gaussian distribution**.

Normal PDF's for $\mu = 0$ and $\sigma = 0.5, 1, 1.5, 2$.



Normal PDF's for $\sigma = 1$ and $\mu = -1, 0, 1$.



Definition 5.16. Normal CDF

The CDF associated with the standard normal distribution is denoted Φ and is given by

$$\Phi(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt \qquad \text{for } x \in \mathbb{R}.$$

There is no simpler expression for $\Phi(x)$ so that tables have to be used in order to evaluate it at $x \neq 0$.

Note that $\Phi(0) = 1/2$.

Proposition 5.17. CDF of Normal RV's

The CDF of $X \sim N(\mu, \sigma^2)$ is given by

$$F_X(x) = \Phi\left(\frac{x-\mu}{\sigma}\right) \qquad \text{for } x \in \mathbb{R}.$$

In this setup, note that $F_X(\mu) = 1/2$.

Example 5.13. Hens and eggs

It is known that hens begin to lay eggs at about 6 months of age. However, younger hens tend to lay smaller eggs, often weighing less than the minimum marketable weight of 54 grams. It is believed that the weight of eggs produced by young hens is normally distributed with $\mu = 50.3$ grams and $\sigma = 5.3$ grams.

Which proportion of eggs laid by young hens exceed the minimum weight of 54 grams?

After reaching 1 year of age, hens produce eggs whose weight is normally distributed with $\mu = 67.1$ grams and $\sigma = 6.2$ grams.

Find the proportion of eggs laid by older hens with a weight that is above the required 54 grams.

Example 5.14. Cutoffs

A prof gives a 100-point exam for which she expects the students' scores to be approximately normally distributed with $\mu = 60$ and $\sigma = 10$.

Find the approximate proportion of students taking the exam that will get a mark

- between 55 and 65,
- between 60 and 80.

For this exam, the instructor wants to give an A to 10% of the students, a B to 20%, a C to 30%, a D to 20% and an F to the remaining 20%.

What are the cutoffs that she should use?

Example 5.15. The 68/95/99.7 rule revisited

Let $X \sim N(\mu, \sigma^2)$.

Find $\mathbb{P}(|X - \mu| \leq k\sigma)$ for $k = 1, 2, 3$.

Example 5.16. Standardized normal random variable

Assume that $X \sim N(\mu, \sigma^2)$ and let $Y = (X - \mu)/\sigma$.

Show that Y has the standard normal distribution.

Theorem 5.18. Local Form of the DeMoivre-Laplace Limit Theorem

Let $X \sim \text{Bin}(n, p)$. Then , we have that

$$\mathbb{P}(X = x) \simeq \frac{1}{\sqrt{2\pi}\sqrt{np(1-p)}} e^{-(x-np)^2/2np(1-p)} \qquad \text{for } x = 0, 1, \ldots, n.$$

In other words, we have that

$$P_X(x) \simeq f(x),$$

where f corresponds to the PDF of the $N(np, np(1-p))$ distribution.

The approximation is good when n is large and p is not too close to 0 or 1.

► Notes:

- Abraham DeMoivre first proved this in 1733 for $p = 1/2$.
- Pierre Laplace then extended the result to $p \neq 1/2$ in 1812.
- It is the work of Karl Friedrich Gauss (1777-1855) that established the usefulness and importance of normal distributions.
- See Figure 8.16 from the textbook!

Example 5.17. Coin flipping

Flipping a fair coin 20 times, calculate the exact probabilities of observing 4, 7 and 10 results of *Heads* and their normal approximation.

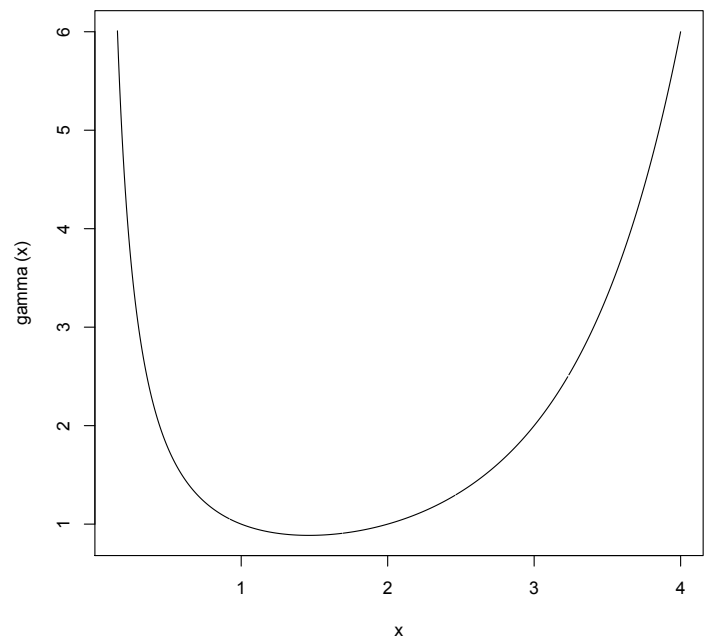
5.4.4 Gamma RV's (*Weiss §8.6*)

Definition 5.19. The Gamma Function

The **gamma function**, denoted Γ, is defined as

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt \qquad \text{for } x > 0.$$

Gamma function: $\Gamma(x)$



Proposition 5.20. Properties of the Gamma Function

The gamma function has the following properties.

1. $\Gamma(x+1) = x\,\Gamma(x) \qquad \text{for } x > 0.$
2. $\Gamma(n) = (n-1)! \qquad \text{for } n = 1, 2, \ldots$
3. $\Gamma(1/2) = \sqrt{\pi}.$

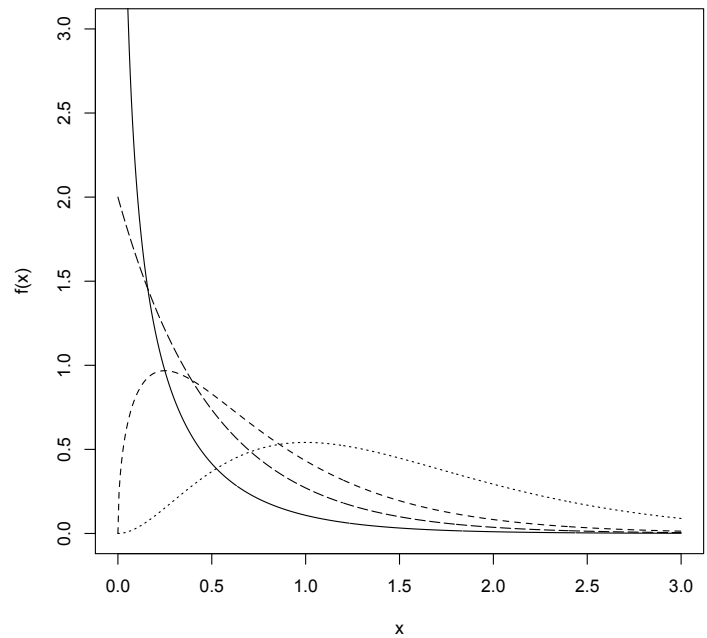
Definition 5.21. Gamma Random Variable

A continuous random variable X is said to have the **gamma distribution** with parameters $\alpha > 0$ and $\lambda > 0$ if its PDF is given by

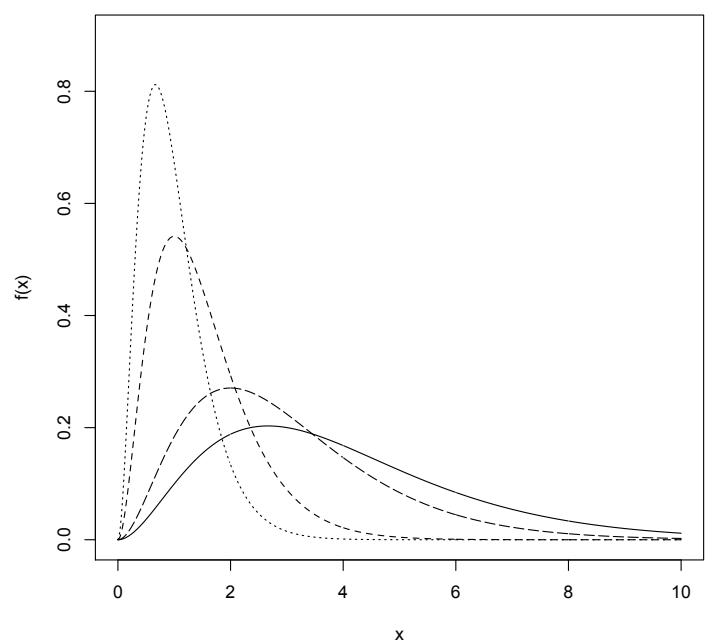
$$f_X(x) = \begin{cases} \dfrac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} & \text{for } x > 0, \\ 0 & \text{otherwise.} \end{cases}$$

We write $X \sim \Gamma(\alpha, \lambda)$ or $X \sim \text{Gamma}(\alpha, \lambda)$.

Gamma PDF's with $\lambda = 2$ and $\alpha = 1/2, 1, 3/2, 3$.



Gamma PDF's with $\alpha = 3$ and $\lambda = 3/4, 1, 2, 3$.



Example 5.18. Link between gamma and Poisson distributions

For any $t > 0$, define $N(t)$ as the random number of occurrences of some event over the time interval $[0, t]$. In addition, assume that

$$N(t) \sim \text{Poisson}(\lambda t),$$

for some $\lambda > 0$. Show that the time T of occurrence of the nth event (or, waiting time until the nth occurrence of the event) has a gamma distribution with parameters n and λ.

Example 5.19. A gamma density

A random variable Y has the following PDF

$$f_Y(y) = \begin{cases} cy^5 10^{-y} & \text{for } y > 0, \\ 0 & \text{otherwise.} \end{cases}$$

Find the value of c. Also, find the mode of this density function, that is, the point y at which the density attains its maximum.

Hint: if g is a positive function, then $g(x)$ and $\ln[g(x)]$ achieve their maximum at the same point.

Example 5.20. (refer to Example 5.12)

At Luigi's Pizzeria, orders of pizza are made with rate $\lambda = 0.2$ orders per minute (that is, one order every 5 minutes).

Find the probability that at least three orders are made in the next 15 minutes.

5.4.5 Beta RV's (*Weiss §8.6*)

Definition 5.22. The Beta Function

The **beta function**, denoted B, is defined as

$$B(x, y) = \int_0^1 t^{x-1}(1-t)^{y-1} dt \qquad \text{for } x > 0, y > 0.$$

Note that the beta function is related to the gamma function through

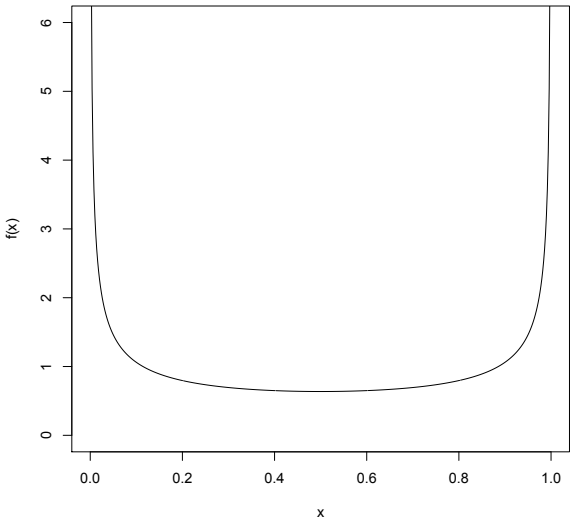
$$B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}.$$

Definition 5.23. Beta Random Variable

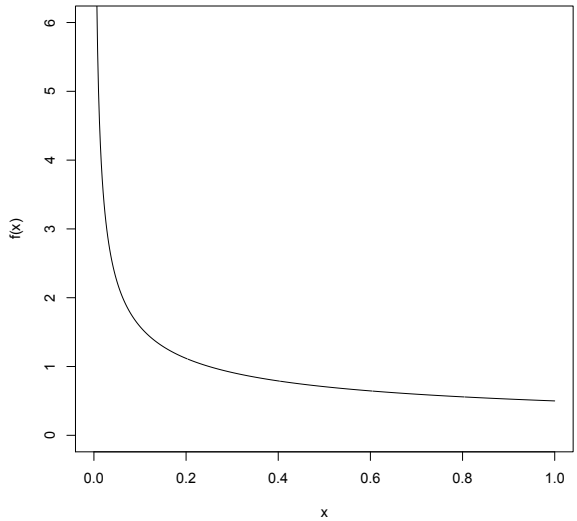
A continuous random variable X is said to have the **beta distribution** with parameters $a > 0$ and $b > 0$ if its PDF is given by

$$f_X(x) = \begin{cases} \dfrac{1}{B(a,b)} x^{a-1}(1-x)^{b-1} & \text{for } 0 < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

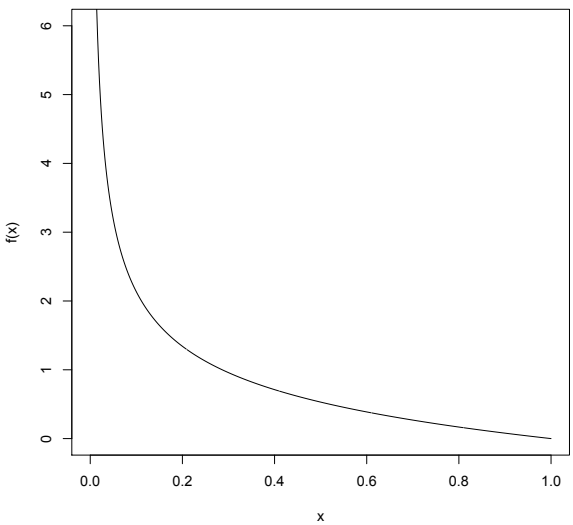
We write $X \sim \mathcal{B}(a, b)$ or $X \sim \text{Beta}(a, b)$.



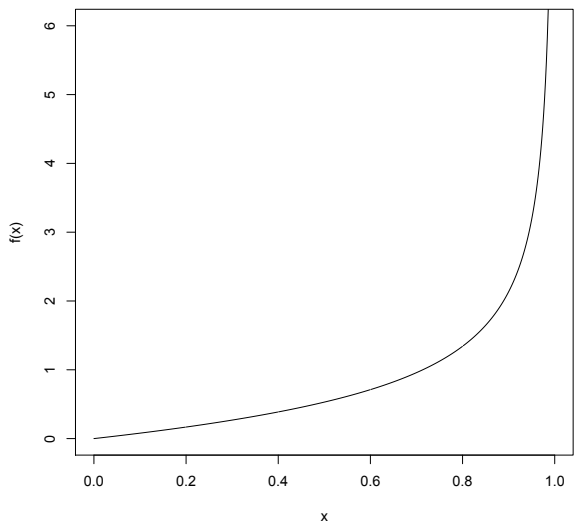
$a = 1/2, b = 1/2$



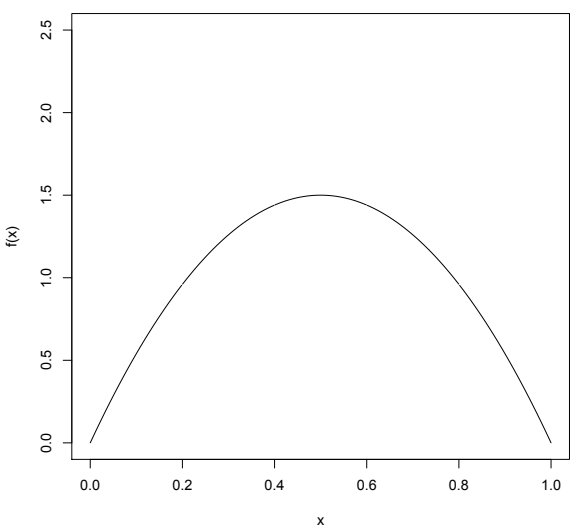
$a = 1/2, b = 1$



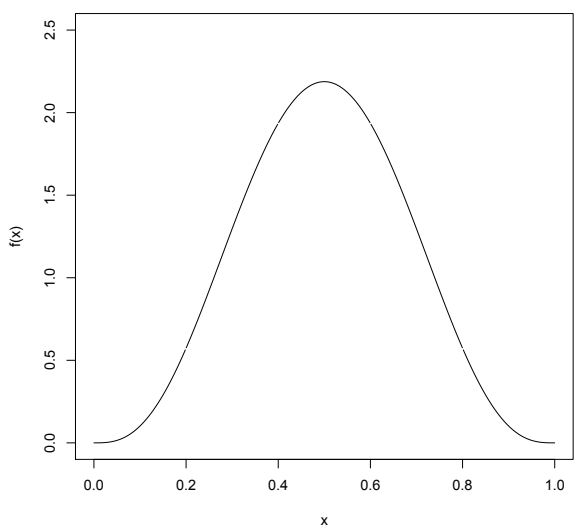
$a = 1/2, b = 2$



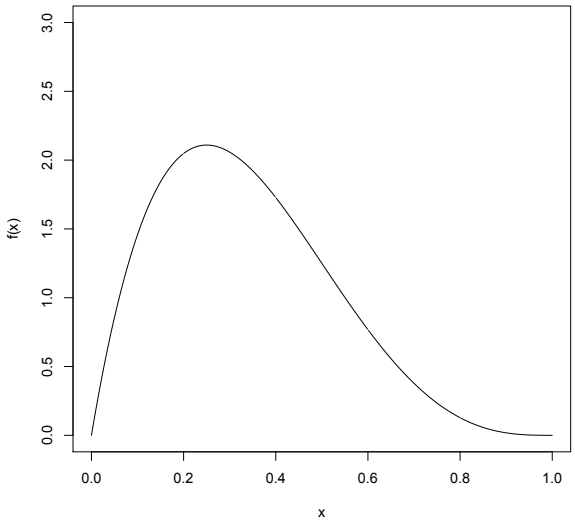
$a = 2, b = 1/2$



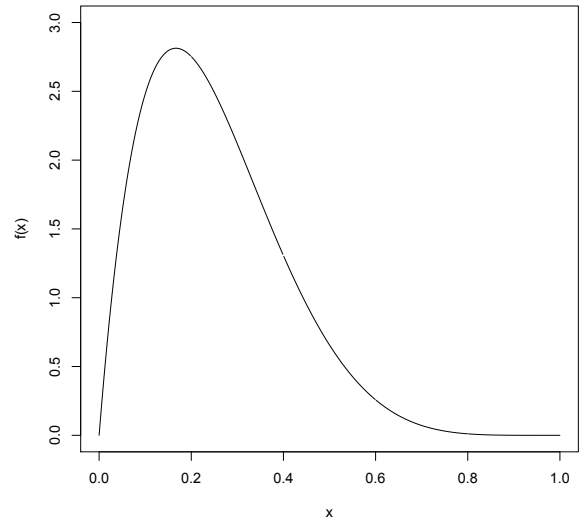
$a = 2, b = 2$



$a = 4, b = 4$



$a = 2, b = 4$



$a = 2, b = 6$

Example 5.21. Two beta densities

The random variables X and Y have the following PDF's:

$$f_X(x) = \begin{cases} c_1 x(1-x)^3 & \text{for } 0 < x < 1, \\ 0 & \text{otherwise,} \end{cases} \qquad \text{and} \qquad f_Y(y) = \begin{cases} c_2(1/\sqrt{y} - \sqrt{y}) & \text{for } 0 < y < 1, \\ 0 & \text{otherwise.} \end{cases}$$

Find the value of both constants c_1 and c_2. Also find the CDF's of X and Y.

5.5 Functions of a Continuous RV (*Weiss §8.7*)

Finding the distribution of $Y = g(X)$: the CDF method

Assume that X is a continuous RV with known PDF or CDF, and that $Y = g(X)$ where g is a real-valued function that is defined and differentiable on R_X, the range of X.

To find the CDF or a PDF of Y:

1. identify the CDF or PDF of X,
2. identify the range R_Y of Y,
3. for $y \in R_Y$, find $F_Y(y)$ in terms of F_X,
4. if needed, f_Y is obtained by differentiating F_Y.

Example 5.22. Link between uniform and exponential distributions

Assume that $X \sim \text{Unif}(0, 1)$. Find the distribution of $Y = -\ln X$.

Example 5.23. The absolute value
Let $X \sim \text{Unif}(-1, 1)$ and let $Y = |X|$.
Show that $Y \sim \text{Unif}(0, 1)$.

Proposition 5.24. Linear Transformation of a Normal RV
Assume that $X \sim N(\mu, \sigma^2)$ and let $Y = a + bX$ with $b \neq 0$.
Then, we have that $Y \sim N(a + b\mu, b^2\sigma^2)$.

Example 5.24. Playing with normal RV's
Let $X \sim N(2, 8)$.
What are the distributions of $Y_1 = 1 + 2X$ and $Y_2 = 1 - 2X$?
Also, find the constants a and b such that $U = a + bX \sim N(10, 2)$.

Theorem 5.25. Univariate Transformation Theorem
Let X be a continuous random variable with PDF f_X and let g be a real-valued function defined on R_X. If g is strictly monotone (*i.e.* increasing or decreasing) and differentiable on R_X, then the PDF of $Y = g(X)$ is

$$f_Y(y) = \begin{cases} \dfrac{f_X(x)}{|g'(x)|} & \text{for } y \in R_Y, \\ 0 & \text{otherwise,} \end{cases}$$

where $x = g^{-1}(y)$ and $R_Y = \{y : y = g(x) \text{ for some } x \in R_X\}$.

Finding the distribution of $Y = g(X)$: using the Transformation Theorem
Assume that X is a continuous RV with known PDF or CDF, and that $Y = g(X)$ where g is a real-valued function that is defined and differentiable on R_X, the range of X.
To find a PDF of Y:

1. identify the PDF of X and check that g is strictly monotone,
   (if not, use CDF method)
2. identify the range R_Y of Y,
3. for $y \in R_Y$, find $f_Y(y)$ in terms of x using the theorem,
4. for $y \in R_Y$, solve for $x = g^{-1}(y)$,
5. f_Y is obtained from 3 and 4.

Example 5.25. (refer to Example 5.22)

Assume that $X \sim \text{Unif}(0, 1)$.

Find the distribution of $Y = -\ln X$ using the Univariate Transformation Theorem.

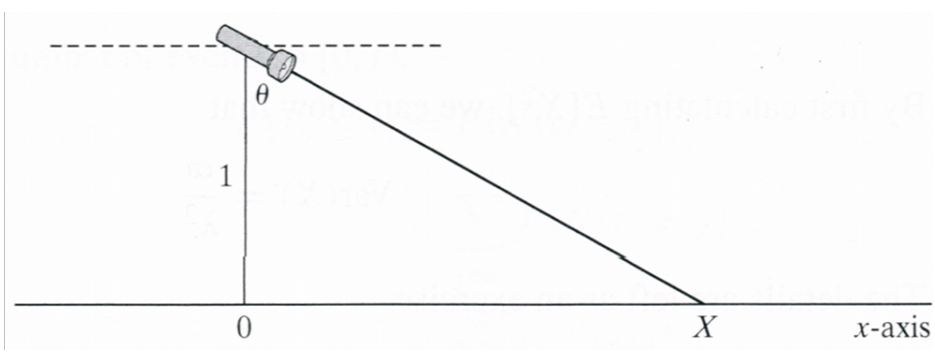
Example 5.26. Can the Univariate Transformation Theorem be used?

In which of the following setups can the Univariate Transformation Theorem be used to obtain the distribution of Y?

- $X \sim \text{Unif}(-1, 1)$ and $Y = |X|$
- $X \sim \text{Exp}(3)$ and $Y = X^2$
- $X \sim N(0, 8)$ and $Y = X^2$
- $X \sim N(0, 8)$ and $Y = X^3$

Example 5.27. The Cauchy distribution

Suppose that a narrow beam flashlight is spun around its center as in the figure below. When the flashlight has stopped spinning, consider the point X at which the beam intersects the x-axis. Note that the point X can be determined from the angle θ between the flashlight and the y-axis. It is further assumed that $\theta \sim \text{Unif}(-\pi/2, \pi/2)$.



Find the PDF of X.